

CUSTOMER MILESTONE BRIEF

MS #30 - Customer Source Analysis Workcell

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.30.0
Date	2026-08-29
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #30 - Customer Source Analysis Workcell converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Builds a customer-specific graph and analysis receipt rather than presenting the CardDemo reference graph as customer evidence.

Why this matters to a customer

Builds a customer-specific graph and analysis receipt rather than presenting the CardDemo reference graph as customer evidence.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a deterministic customer source analysis workcell bound to each approved intake rather than projecting the repository's CardDemo graph as customer evidence.
- Added a dedicated content-addressed relationship ontology spanning bounded COBOL, copybook, PL/I, JCL, Db2 DDL/SQL, HLASM control flow, IMS DBD/PSB structure, VSAM IDCAMS definitions, and approved CICS configuration relationships.
- Added customer-specific gzip graph snapshots and analysis receipts with parser coverage, entity counts, graph statistics, exact source identities, and visible unresolved or ambiguous references.
- Added native COBOL-to-PL/I calls, JCL-to-PL/I/HLASM execution, HLASM branch targets, IMS DBD-to-segment and PSB-to-PCB lineage, and VSAM cluster-to-AIX-to-path relationships inside the pilot analysis contract without

changing the canonical CardDemo graph identity.

- Added fail-closed source re-hashing, graph/ontology validation, size bounds, analysis tamper tests, and explicit rejection of behavior, live-system, mainframe-equivalence, and production claims.
- Added a v2 pilot dossier that binds the customer analysis while preserving the v1 intake, preflight, and analysis contracts and the historical v1 dossier schema.
- Extended POSIX and PowerShell pilot flows with an explicit analyze stage and made the six-artifact, nine-source-class reference rehearsal byte-reproducible on Linux, macOS, and Windows.

How it differs from earlier work

MS #30 builds on MS #29 (Governed Source-Only Pilot Intake). The earlier milestone established its own bounded capability; this milestone adds customer source analysis workcell while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use customer source analysis workcell as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.30.0
- README.md product and operating guidance

- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.